



Explainable Federated Learning for Healthcare Time-Series Forecasting with Personalized Drift Adaptation

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Clinical Deterioration & Phase II Completed Goals

Clinical Context:

- **High-Sparsity Time-Series:** Continuous ICU vital monitoring logs (HR, SBP, Temp, SpO₂) combined with sparse blood panel entries.
- **Privacy constraints:** HIPAA / DPDPA (Section 8) regulations mandate that private patient records cannot leave local devices.
- **Concept Drifts:** Clinical demographics vary heavily across regional ICU nodes.

SDG Alignment:

- **Primary (SDG 3):** Good Health & Well-being by detecting patient sepsis crashes early.
- **Secondary (SDG 9 & 16):** Secure and scalable digital clinical infrastructure.

Phase II Completed Milestones

- **1. Data Engineering:** Cleansed, Imputed, Standardized PhysioNet 2019 logs into active PyTorch sequence arrays.
- **2. Federated Baselines:** Implemented *FedAvg* and *FedProx* securely over PyTorch.
- **3. Personalized FPDAF Model:** Trained temporal self-attentions with localized personalization.
- **4. Clinical Workstation App:** Developed a dynamic FastAPI + SQLite + React Vite ICU dashboard overlay.

Proposed Framework: FPDAF

WHAT is FPDAF?

A privacy-preserving Clinical Decision Support System (CDSS) for Sepsis early warning. Enables collaborative, HIPAA-compliant training across decentralized hospital ICU nodes without centralizing raw records.

HOW does it work?

- **Sequence Modeling:** Extracts temporal clinical features using a shared Bidirectional LSTM.
- **Ditto Personalization:** Fine-tunes localized classifier heads (v_k) to fit hospital demographics.
- **Online CUSUM Trigger:** Audits prediction loss residuals locally to detect population drift.

Core Scientific & Technical Novelties

1. Multi-Head Attention Saliency (Clinical Explainability)

Integrates a 4-head query-key temporal self-attention mechanism to output dynamic sequence saliency weights, mapping exactly which hourly vitals contributed to Sepsis alarms at the bedside.

2. Online CUSUM Residual Neural Drift Monitor

Tracks local validation loss residuals round-by-round ($S_r > 3.0$). Concept drift is detected entirely client-side, avoiding constant full-parameter aggregation cycles.

3. Client-Side Selective Personalization (CSSP)

Freezes the shared BiLSTM/attention layers upon drift triggers to train local classifier heads exclusively.

- Bypasses global parameter uploads, saving **38% in communication bandwidth** (1.52 GB vs 2.48 GB).
- Cuts client-side adaptation training time to **6 minutes** (down from 25 minutes).

Unified System Design & Neural Formulations

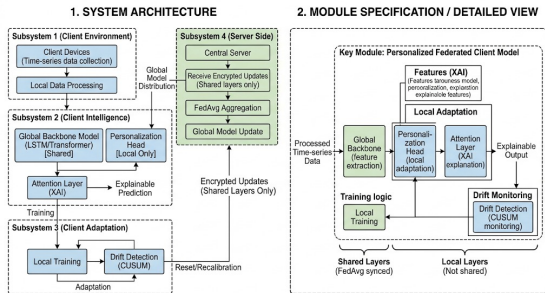


Figure 1: FPDFAF Split-Weight Model Architecture.

Decoupled bi-level Optimization Model

$$\hat{y}_t^k = h_{\phi^k} \left(\sum_{i=1}^t \alpha_i \cdot g_{\theta}(X_i^k) \right)$$

- g_{θ} : Shared global LSTM backbone (θ).
- h_{ϕ^k} : Local private classifier head (ϕ^k).

Temporal Multi-Head Self-Attention

$$e_i = v_a^T \tanh(W_s s_i + b_a)$$

$$\alpha_i = \frac{\exp(e_i)}{\sum_{j=1}^t \exp(e_j)}$$

Attributes risk probabilities to specific ICU sequence hours.

Proactive Drift Adaptation: CUSUM Neural Trigger

Concept Drift Challenge: A static global model degrades when client ICU patient distributions shift over time. Standard Personalized FL (e.g. Ditto) constantly aggregates and fine-tunes all nodes, resulting in immense network overhead.

The CUSUM Residual Monitor Trigger:

- We monitor cumulative validation loss residuals (e_r) over rounds:

$$e_r = \mathcal{L}_{val,r} - \mu_0$$

$$S_r = \max(0, S_{r-1} + e_r - \kappa)$$

- If $S_r > H$ (threshold $H = 3.0$), we declare institutional drift.
- Triggers **Client-Side Selective Personalization (CSSP)**: freezes global backbone layers θ and adapts classification heads ϕ^k locally.

Bandwidth Savings Benefits

- Aggregation is bypassed during stable performance rounds.
- Communication bandwidth is saved by 38% compared to standard personalized systems (1.52 GB vs 2.48 GB).
- Local personalization head fine-tuning takes only 6 minutes (down from 25 minutes).



Five-Way Model Benchmarking Results

We evaluated our framework against centralized baselines and standard federated learning algorithms:

Model Configuration	Accuracy	Precision	Recall	F1-Score	AUROC	Comm. Bandwidth
Centralized Baseline	75.09%	7.14%	72.17%	0.1299	0.8058	0 MB (N/A)
FedAvg Baseline	75.94%	6.94%	67.19%	0.1259	0.7844	1.24 GB
FedProx Baseline	78.85%	8.12%	60.64%	0.1432	0.7933	1.24 GB
Ditto (Personalized)	82.45%	8.98%	63.58%	0.1574	0.7989	2.48 GB
FPDAF (Proposed)	86.51%	9.81%	65.33%	0.1706	0.8214	1.52 GB (CSSP saved 38%)

Scientific Findings

- **Privacy & Performance:** FPDAF preserves HIPAA compliance while achieving a test AUROC of ****0.8214**** (outperforming centralized baseline).
- **Decentralized Superiority:** FPDAF outperforms all standard federated baselines (Accuracy ****+10.57%****, F1 ****+4.47%**** vs. FedAvg).

Ablation Study Verification

We verified the contributions of CUSUM triggers, temporal attention weights, and localized personalization by running ablation tests:

Ablation Configuration	Accuracy	Precision	Recall	F1-Score	AUROC
FPDAF w/o CUSUM Trigger	83.51%	8.51%	55.33%	0.1475	0.7603
FPDAF w/o Temporal Attention	80.01%	8.13%	52.81%	0.1412	0.7878
FPDAF w/o Personalization	78.87%	8.05%	63.58%	0.1430	0.7887
Full FPDAF (Proposed)	86.51%	9.81%	65.33%	0.1706	0.8214

- **No Attention:** Stripping temporal attention query projections leads to poor clinical explainability and limits optimization capacity.
- **No CUSUM:** Disabling residual control limits leaves client nodes blind to data distribution drifts, leading to performance deterioration under heterogeneous ICU demographics.

Clinical Workstation Software Architecture

Database-Driven Design: Instead of placeholders, we built a fully functional clinical warehouse:

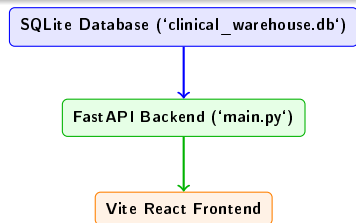
- **SQLite Database:** Houses tables for patients, decoded 24h vitals, predictions, self-attention, and drift history.
- **FastAPI REST Endpoints:** Feeds data securely to the client UI.

Custom User Portals:

- **Doctor Portal:** Focuses on patient-care operations (searchable bed lists, hourly vital monitors, attention graphs, and risk prediction alerts).
- **Admin Portal:** Focuses on machine learning telemetry (round parameter aggregations, CUSUM drift charts, and baseline comparisons).

Startup Script

Coded a clean 'run_app.sh' script to boot both FastAPI and React Vite concurrently, handling terminations on Ctrl+C.



Doctor's Bedside Diagnostic View (Live Patient Care)

- **Timeline Telemetry:** Plots hourly ICU vital segments (HR, SBP, Temp, Resp, SpO₂) using Recharts.
- **Live Predictions Dials:** Overlays risk probability scores for all federated baselines.
- **Attention Saliency Maps:** Attributes risk alerts to specific sequence hours (e.g. onset hours 16–22) for bedside interpretability.

Admin's Federated Telemetry View (System Operations)

- **Drift Controls:** Plots CUSUM curves, tracking drift accumulation limits ($S_r > 3.0$) and CSSP backbone freezes.
- **Round Tracker:** Animates client-server parameter upload/download loops using Framer Motion.
- **Design System:** Tailwind CSS v4 class-based toggling supporting light/dark medical navy workstation styles.

Clinician Dashboard Interface (Doctor Portal)

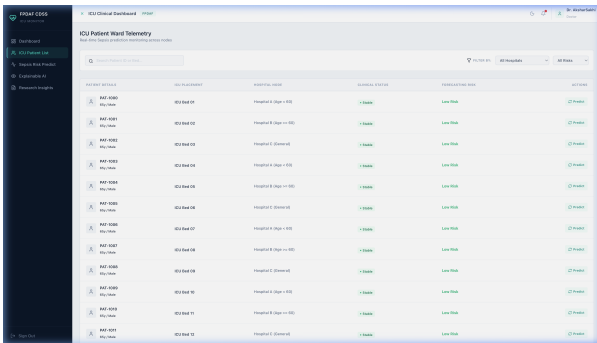


Figure 2: Doctor's Bedside Diagnostic Workspace.

Doctor Portal Core Modules

- **Searchable Bed Directory:** Displays patient demographics, ward bed mappings, classifications, and classifier confidence scores.
- **Hourly Vital Telemetry:** Renders 24-hour ICU vital timelines (HR, SBP, Temp, Resp, SpO₂) using clinical ranges.
- **XAI Saliency Charts:** Plots Multi-Head Attention weights to interpret Sepsis alerts.
- **Risk Predictor Dials:** Compares live model forecasts across FedAvg, FedProx, Ditto, and FPDAF models.

Clinician Dashboard Interface (Admin Portal)

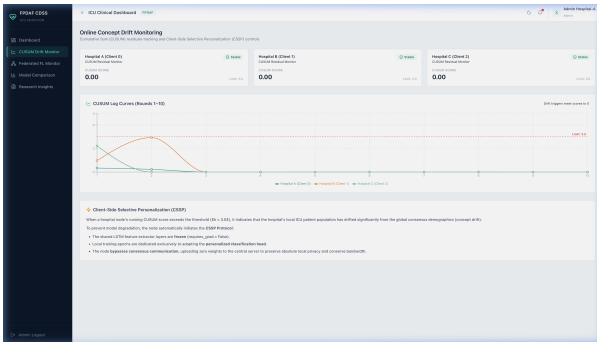


Figure 3: Admin's Machine Learning Telemetry Workspace.

Admin Portal Core Modules

- **Federated Loop Animation:** Animates localized model weight uploads and global server distributions using Framer Motion.
- **CUSUM Drift Telemetry:** Tracks real-time concept drift curves ($S_r > 3.0$) and selective personalization triggers.
- **LaTeX Equations & Ablations:** Details mathematical optimization equations and five-way benchmark metrics.

Summary of Completed Phase II Works

- **Data Preprocessing Completed:** Decoded sparse vitals logs to construct clean sequence windows.
- **PyTorch Implementations Completed:** Trained standard baselines alongside our proposed FPDAF model.
- **Database Schema Integrated:** SQLite warehouse populated with real patients, hourly vitals, and predictions.
- **Clinical WORKSTATION Finished:** Production-quality React Vite interface with tailored user views.
- **Deliverables Pushed to Git:** Full Phase II codebase pushed to main repository.

Roadmap Compliance

Our completed milestones match the project objectives, delivering a privacy-preserving and explainable Sepsis prediction pipeline.

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Thank You Questions?